Testing

Automated test suite for the five P0 stories of PriceLens. Conventions for writing tests and editing this document are in testing-guidelines.md.

1. What we test, and why

Layer	Where	How we test it	Why
Services	`backend/src/modules/**/*.service.js`, `backend/src/background/**`	Unit tests that call exported functions directly with plain arguments	Every business rule (BR1, BR2, BR4, BR5, BR6, BR8) is decided here, so a mistake in this layer gives users a wrong price, alert or notification
Repository	`backend/src/modules/**/*.repository.js`	Run through the service tests against a fake database client that answers each query; no PostgreSQL	Checks the values each rule sends to the database (variant, currency, comparison operator) and how rows are mapped, without needing a database on the test machine
Routes	`backend/src/modules/**/*.routes.js`	Not tested in this suite	Routes only read the request and turn a service result or error into an HTTP response; they hold no business rule
Interface	`frontend/src/**`	Not tested in this suite	Checked by hand against every P0 acceptance criterion in the browser; the screens still change often, so automated UI tests would break on layout changes rather than on rule changes

Where the risk is

We tested the alert and notification rules most (US03 and US05): they contain the only numeric limits that decide whether a user is refused or sent an email, so an off-by-one mistake (`<` instead of `<=`, 20 instead of 19) goes unnoticed in a demo but misleads users. Every numeric rule is therefore tested on both sides of its limit, and the comparison operators written in SQL are read and evaluated by the fake database so that changing them turns a test red.

Known gaps

- **US02:** the date and `IN_STOCK` filtering of price history runs in SQL (`price-history.repository.js`). Because the suite runs without a database, it checks that the 30-day UTC window is computed correctly and how rows are assembled, but not the SQL query itself.
- **US03:** alert creation runs inside a database transaction, so the tests replace the database connection with a fake client. BR2 price comparisons are written as PostgreSQL `NUMERIC` expressions; the fake evaluates the same comparison operator the service sends, so a change from `<` to `<=` is caught, but PostgreSQL's own decimal arithmetic is not exercised.
- **US04:** duplicate URLs (BR6) are rejected by a unique constraint in PostgreSQL, not by application code. The tests simulate the database's duplicate-key error and check that it becomes the "already being tracked" response; the constraint itself is not exercised.
- **US05:** the tests cover how a new price creates a pending notification (BR4 and the subject line). Sending the email within 5 minutes depends on the delivery job, the real clock and PostgreSQL locks, so it is not checked here; it is verified in `docs/p0-acceptance.md`. The fake database evaluates the same threshold comparisons the processor sends to PostgreSQL.
- **US09:** choosing each retailer's latest price, sorting offers (in stock first, then by price) and computing the difference from the lowest run in SQL (`offer-comparison.repository.js`). The fake database returns rows in that order and filters them by the variant and currency the service sends, so BR5 and the BR8 status rules are checked, but the SQL sorting and arithmetic are not.

2. Test cases for every P0 story

Each story has at least one `normal` case that succeeds, at least one `error` case where a business rule refuses the input and, when its rule contains a number, boundary cases on both sides of that number. Every ID below matches a test with the same ID in `tests/`.

US02 — View product price history

US02 has no numbered business rule. Its numeric limit is the chart threshold: a chart needs at least 2 valid daily prices.

ID	Story	Input	Expected	Kind
TC-US02-01	US02 View price history	30 valid daily prices, viewed at 2026-10-20 12:00 UTC (AC1)	`AVAILABLE`; exactly 30 points; window 2026-09-20 → 2026-10-19 UTC	normal
TC-US02-02	US02 View price history	Lowest 18,500,000 VND, highest 20,000,000 VND, current 18,900,000 VND (AC2)	Summary shows exactly 18,500,000 / 20,000,000 / 18,900,000 VND	normal
TC-US02-03	US02 View price history	No valid price records (AC4)	`NO_HISTORY`; "No price history available"; no points, no summary	error
TC-US02-04	US02 View price history	Request the history of a product the user does not own	Refused: 404 `TRACKED_PRODUCT_NOT_FOUND`, "Tracked product not found"	error

ID	Story	Input	Expected	Kind
TC-US02-05	US02 View price history	1 valid price record (AC3)	`INSUFFICIENT_DATA`; "Insufficient data for a 30-day chart"	boundary, just below
TC-US02-06	US02 View price history	2 valid price records	`AVAILABLE`; 2 points	boundary, on the limit

US03 — Create a price alert

BR1 limits a user to 20 active alerts. BR2 requires a target price above 0 and strictly below the current price. All cases use a current price of 15,000,000 VND.

ID	Story	Input	Expected	Kind
TC-US03-01	US03 Create a price alert	Target 14,000,000 VND (AC1)	Alert saved as `ACTIVE` with target 14,000,000 and current 15,000,000 VND	normal
TC-US03-02	US03 Create a price alert	Target 16,000,000 VND (AC2)	Refused: 422 "Target price must be lower than the current price"; nothing saved (BR2)	error
TC-US03-03	US03 Create a price alert	Target 0 VND (AC3)	Refused: 422 "Target price must be greater than 0"; nothing saved (BR2)	error
TC-US03-04	US03 Create a price alert	User already has 19 active alerts	20th alert saved as `ACTIVE` (BR1)	boundary, on the limit
TC-US03-05	US03 Create a price alert	User already has 20 active alerts (AC4)	Refused: 409 "Maximum 20 active alerts reached"; nothing saved (BR1)	boundary, just above
TC-US03-06	US03 Create a price alert	Target 15,000,000 VND, equal to the current price	Refused: 422 "Target price must be lower than the current price"; nothing saved (BR2)	boundary, on the limit

US04 — Add a product using its URL

BR6 allows one watchlist record per normalized product URL; tracking parameters such as `utm_source` are removed before the duplicate check. BR6 has no numeric limit, so US04 has no boundary cases. All cases use the iPhone 15 128GB Black offers from the demo data.

ID	Story	Input	Expected	Kind
TC-US04-01	US04 Add a product	`https://digital-hub.example/apple-iphone-15-128gb-black` (AC1)	Resolved to Digital Hub, Apple iPhone 15, with variant iPhone 15 128GB Black to select	normal

ID	Story	Input	Expected	Kind
TC-US04-02	US04 Add a product	Selected Digital Hub offer for iPhone 15 128GB Black (AC1)	One record saved with status `TRACKING` and the normalized URL	normal
TC-US04-03	US04 Add a product	`https://techmart.example/apple-iphone-15-128gb-black?utm_source=qa`	Looked up as `https://techmart.example/apple-iphone-15-128gb-black` (BR6)	normal
TC-US04-04	US04 Add a product	Offer whose normalized URL is already in the user's watchlist (AC2)	Refused: 409 "This product URL is already being tracked"; nothing saved (BR6)	error
TC-US04-05	US04 Add a product	`not-a-url` (AC3)	Refused: 422 "Unsupported or invalid product URL"; no lookup made	error
TC-US04-06	US04 Add a product	`https://unsupported.example/product` (AC3)	Refused: 422 "Unsupported or invalid product URL"	error

US05 — Receive a price-drop notification

BR4 sends exactly one notification when a valid price moves from above the target to at or below it, sends nothing while the price stays at or below the target, and re-arms only after the price rises above the target again. All cases use an active alert on the Dell XPS 13 9340 with a target of 14,000,000 VND.

ID	Story	Input	Expected	Kind
TC-US05-01	US05 Receive a notification	Previous price 15,000,000 VND, new price 13,900,000 VND (AC1)	Exactly 1 pending notification for `demo@pricelens.local`; alert marked at or below target (BR4)	normal
TC-US05-02	US05 Receive a notification	Same price drop as TC-US05-01 (AC3)	Subject is exactly "Price Drop Alert: Dell XPS 13 9340"	normal
TC-US05-03	US05 Receive a notification	After a notification at 13,900,000 VND, new price 13,800,000 VND (AC2)	No further notification; alert stays at or below target (BR4)	error
TC-US05-04	US05 Receive a notification	Price rises from 13,800,000 to 14,200,000 VND (AC4)	No notification; alert re-armed as above target (BR4)	normal
TC-US05-05	US05 Receive a notification	Re-armed alert, price falls from 14,200,000 to 14,000,000 VND (AC4)	Exactly 1 new notification (BR4)	boundary, on the limit
TC-US05-	US05 Receive a notification	Re-armed alert, price falls from 14,200,000 to	No notification; alert stays above target (BR4)	boundary, just above

ID	Story	Input	Expected	Kind
06		14,000,001 VND		

US09 — Compare prices across retailers

BR5 compares only offers for the tracked variant; BR8 never picks an out-of-stock offer as the lowest available offer. The numeric limit is the number of in-stock offers: two or more form a multi-source comparison, exactly one shows "No multi-source comparison is currently available". Cases use the iPhone 15 128GB Black fixture unless stated otherwise.

ID	Story	Input	Expected	Kind
TC-US09-01	US09 Compare offers	TechMart 19,900,000 VND and Digital Hub 20,200,000 VND, both in stock (AC1)	`AVAILABLE`; TechMart listed first as the lowest offer; Digital Hub 300,000 VND above the lowest	normal
TC-US09-02	US09 Compare offers	Fixture also has an in-stock iPhone 15 256GB Black offer at 18,500,000 VND (AC2)	Offers are looked up for the 128GB variant only; the 18,500,000 VND offer is not shown (BR5)	error
TC-US09-03	US09 Compare offers	Gadget World 18,200,000 VND out of stock, Digital Hub 18,500,000 VND in stock	Digital Hub is the lowest available offer; Gadget World shown as `OUT_OF_STOCK` with no difference (BR8)	error
TC-US09-04	US09 Compare offers	Gadget World 18,200,000 VND and Digital Hub 18,500,000 VND, both out of stock	`ALL_OUT_OF_STOCK`; "Currently out of stock across all tracked sources"; no lowest offer (BR8)	error
TC-US09-05	US09 Compare offers	Dell XPS 13 9340 with one in-stock offer, TechMart 34,990,000 VND (AC3)	`SINGLE_OFFER`; "No multi-source comparison is currently available"	boundary, just below
TC-US09-06	US09 Compare offers	Two in-stock offers and one out-of-stock offer	`AVAILABLE`; no message	boundary, on the limit

3. Automated tests

Run the whole suite from the repository root:

```
npm test
```

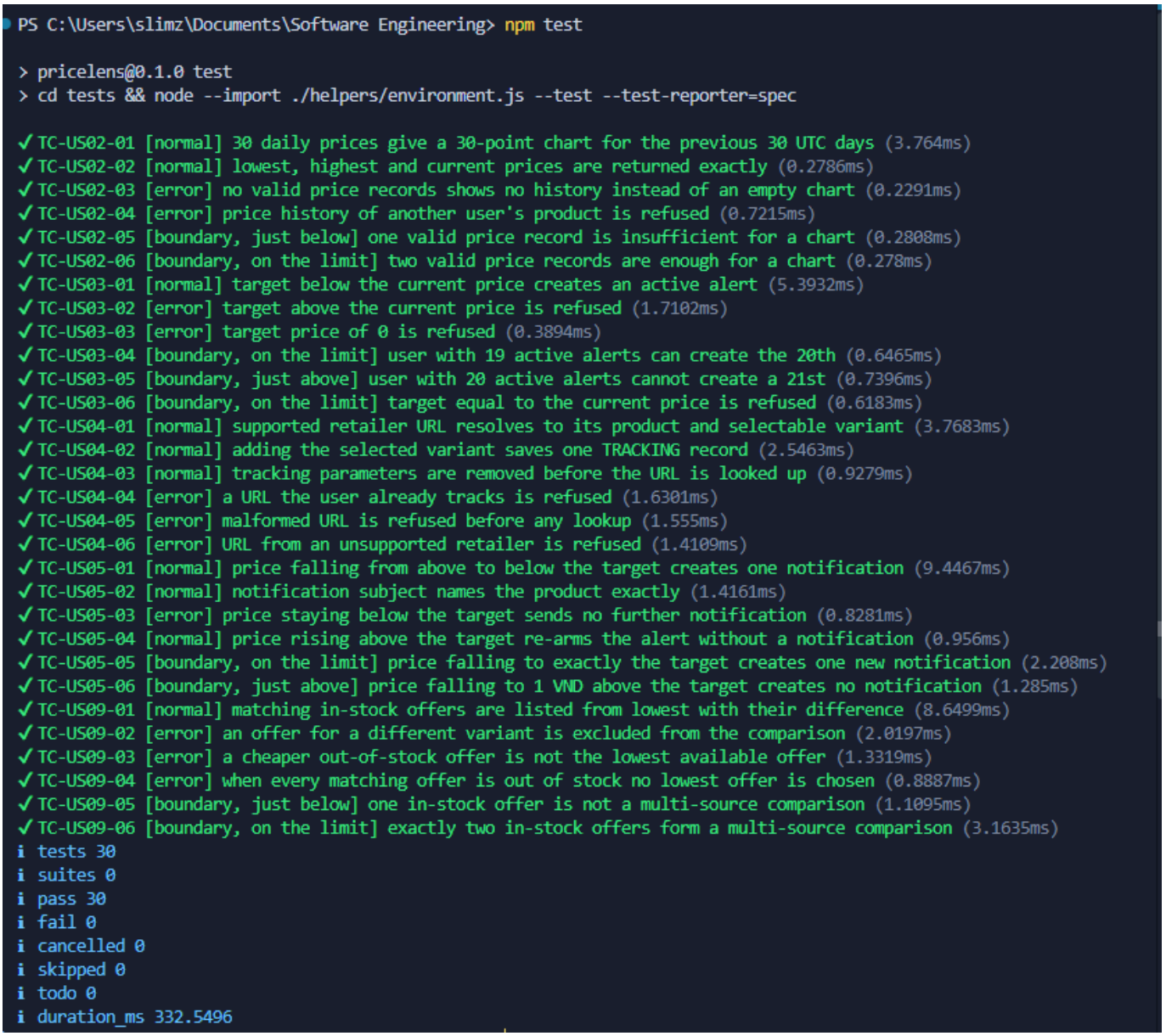
No running server, no real database and no `.env` file are needed; the only prerequisite is the backend dependency install described in SETUP.md.

Story	Test file	Test case IDs
US02	`tests/us02-price-history.test.js`	TC-US02-01 – TC-US02-06

Story	Test file	Test case IDs
US03	`tests/us03-price-alert.test.js`	TC-US03-01 – TC-US03-06
US04	`tests/us04-add-product.test.js`	TC-US04-01 – TC-US04-06
US05	`tests/us05-notification.test.js`	TC-US05-01 – TC-US05-06
US09	`tests/us09-comparison.test.js`	TC-US09-01 – TC-US09-06

Latest run

- **Date:** 2026-11-02
- **Commit:** `39e8722` (`main`)
- **Result:** 30 tests, 30 passed, 0 failed, in under one second



4. What the tests found

The tests found no real bug in the P0 code, so we broke a rule on purpose to show that the tests catch it.

Rule broken: BR1, "a user may have at most 20 active alerts". In `backend/src/modules/tracked-products/price-alert.service.js` we changed the limit check

```
if (activeAlertCount >= MAX_ACTIVE_ALERTS) {
```

to

```
if (activeAlertCount > MAX_ACTIVE_ALERTS) {
```

which lets a user with 20 active alerts create a 21st.

Result: exactly one test turned red, the boundary case just above the limit:

```
✖ TC-US03-05 [boundary, just above] user with 20 active alerts cannot create a 21st
  AssertionError [ERR_ASSERTION]: Missing expected rejection.
    actual: undefined,
    expected: { status: 409, code: 'ACTIVE_ALERT_LIMIT_REACHED', detail: 'Maximum 20
active alerts reached' },

ℹ tests 30
ℹ pass 29
ℹ fail 1
```

TC-US03-04 (19 active alerts, creation allowed) stayed green, which shows the pair of boundary tests pins the limit to exactly 20. We restored `>=` and the full suite passed again (30 of 30).

Every other rule was checked the same way before its tests were merged:

Rule broken on purpose	Test that turned red
US02: chart threshold `pointCount === 1` changed to `<= 2`	TC-US02-06
US03 BR2: `target < current` changed to `<=`	TC-US03-06
US04 BR6: `utm_` parameters no longer removed	TC-US04-03
US05 BR4: `price <= target` changed to `<`	TC-US05-05
US09 BR8: out-of-stock offers counted as available	TC-US09-03, TC-US09-04